

Final Project Proposal

Members

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Overview

Nowadays, most people use online streaming services to listen to music. The project aims to propose a comprehensive data visualization analysis of the most streamed music in 2023, using various musical, temporal, and popularity-based attributes to indicate trends and insights about contemporary song consumption. The project would also examine data across significant platforms such as Spotify, Apple, and Deezer. In addition, we also want to analyze the properties of each song, for example, the energy, valence, bpm, and any other properties of that song. In this project, we would like to use four different data visualization techniques: bar chart, word cloud, bubble chart, and radar chart. First, We use a bar chart to display the rank of songs with the user's selected attributes. As a second feature, we use a word cloud to indicate which artists are trending on the chosen music platform or playlist. Third, we use a bubble chart to analyze when the songs were released and how popular the song is. Also, we can see other properties of the song, like bpm, energy, valence, and so on. Finally, a radar chart is applied to represent the attributes of each song.

Dataset

This dataset contains a comprehensive list of the most famous songs of 2023, as listed on Spotify. The dataset offers many features beyond what is typically available in similar datasets. It provides insights into each song's attributes, popularity, and presence on various music platforms. The dataset includes track name, artist(s) name, release date, Spotify playlists and charts, streaming statistics, Apple Music presence, Deezer presence, Shazam charts, and various audio features.

Attributes:

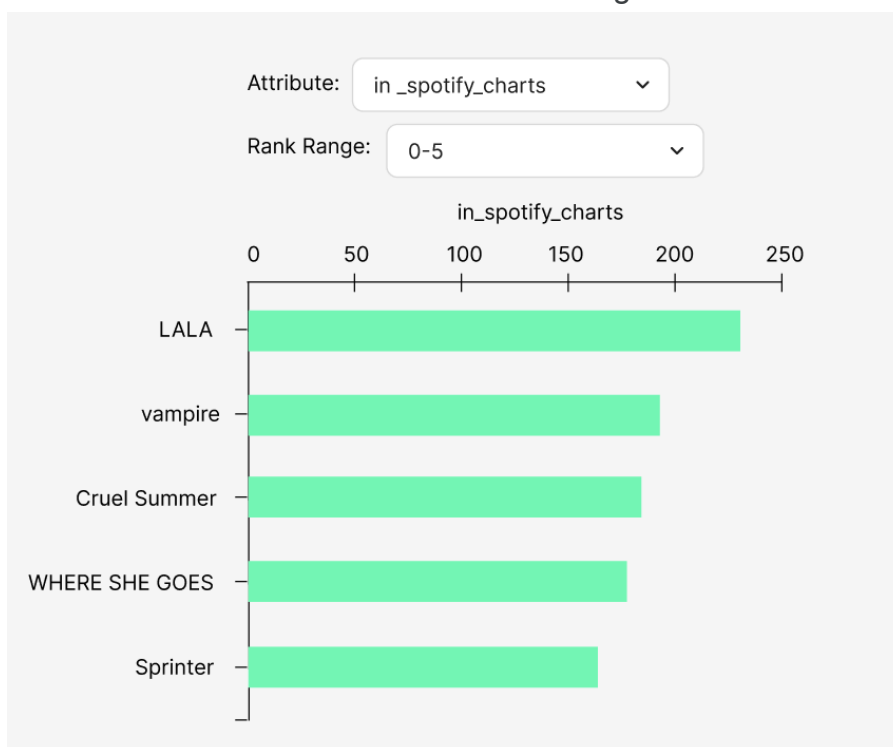
- track_name: *Name of the song*
- artist(s)_name: *Name of the artist(s) of the song*
- artist_count: *Number of artists contributing to the song*
- released_year: *Year when the song was released*
- released_month: *Month when the song was released*
- released_day: *Day of the month when the song was released*
- in_spotify_playlists: *Number of Spotify playlists the song is included in*
- in_spotify_charts: *Presence and rank of the song on Spotify charts*
- streams: *Total number of streams on Spotify*

- *in_apple_playlists*: Number of Apple Music playlists the song is included in
- *in_apple_charts*: Presence and rank of the song on Apple Music charts
- *in_deezer_playlists*: Number of Deezer playlists the song is included in
- *in_deezer_charts*: Presence and rank of the song on Deezer charts
- *in_shazam_charts*: Presence and rank of the song on Shazam charts
- *bpm*: Beats per minute, a measure of song tempo
- *key*: Key of the song
- *mode*: Mode of the song (major or minor)
- *danceability_%*: Percentage indicating how suitable the song is for dancing
- *valence_%*: Positivity of the song's musical content
- *energy_%*: Perceived energy level of the song
- *acousticness_%*: Amount of acoustic sound in the song
- *instrumentalness_%*: Amount of instrumental content in the song
- *liveness_%*: Presence of live performance elements
- *speechiness_%*: Amount of spoken words in the song

Data Visualization

- Bar Chart

We use a bar chart to display the rank of songs. Users can choose the ranking range they want to display. In a bar chart, the y-axis is the song title, and the attributes of the x-axis can be selected by the user. After the user clicks on the song title, he will be directed to the radar chart page, allowing the user to see other information about the song.



(Made by Figma)

- Word Cloud

We use a word cloud to display artist names, with the names of artists who have more streamed songs appearing larger. This makes it easy for users to see which artists are the most popular on a selected music platform or playlist. Users can also choose the platform and playlist they want to explore.



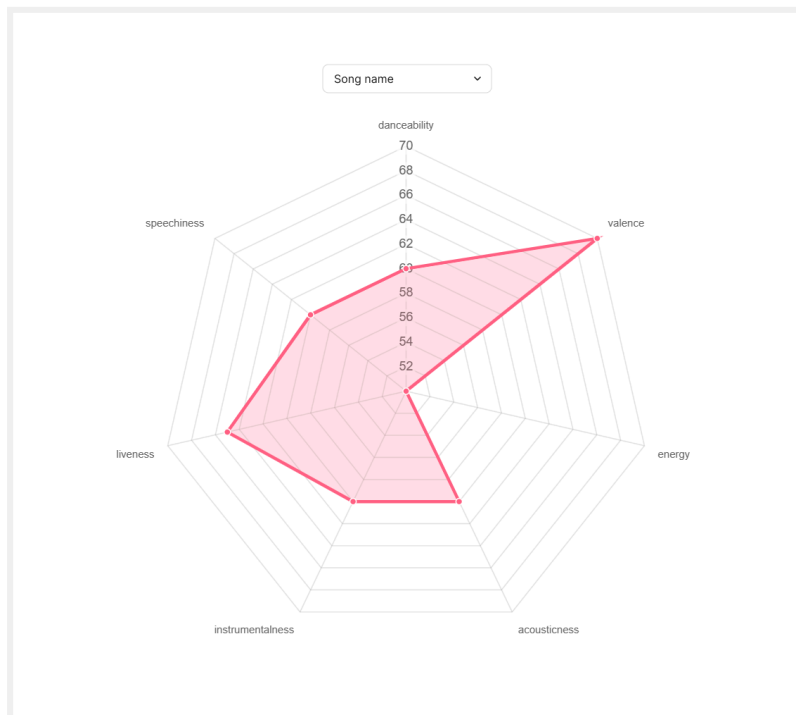
([jasondavies/d3-cloud: Create word clouds in JavaScript.](#))

- Bubble chart

In this part, we would like to use a bubble chart to represent the data. Specifically, we let the x-axis represent the song's released year, and the y-axis represents the month of release. Therefore, the position of the bubbles indicates whether the song is the latest or oldest. On the other hand, the size of the bubble represents the song's streams, valence, danceability, and any other numeric data. By doing so, we can see when the song was released and how popular it is by choosing the bubble size determined by the streams of the songs. If we want to find the latest song, we can visit the upper right corner of the chart; if we see the most energy of the song, we can choose the most significant bubble(choose the energy to decide the bubble size). The diagram is as follows:

- Radar chart

We use a radar chart to visualize each song's attributes, such as danceability, valence, energy, acousticness, instrumentality, liveness, and speechiness. This allows users to quickly understand the characteristics of each song. Also, users can choose the song they want to check.



(Made by Figma)